

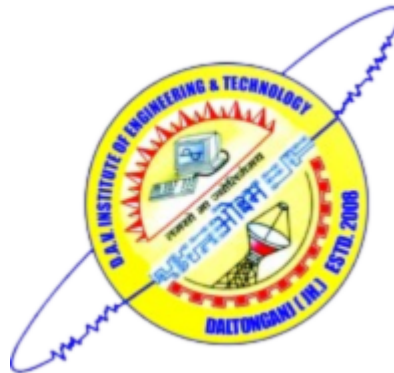
Templify: Cloud-Based Document Management System

A Thesis submitted in partial fulfillment
of the academic requirements for the
degree of Bachelor of Technology in
Computer Science & Engineering

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CERTIFICATE

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I hereby declare that the project report titled “**Templify: Cloud-Based Document Management System**”, submitted in partial fulfillment of the requirements for the degree of **B.Tech in Computer Science & Engineering**, is a record of original work carried out by me under the supervision of **Mr. Rahul Kumar Deo**. This work has not been submitted to any other institution or university for the award of any degree or diploma. In accordance with ethical practices in reporting scientific information, all sources used have been duly acknowledged wherever the findings of others have been cited.

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ACKNOWLEDGEMENTS

I would like to express my sincere gratitude and appreciation to my project supervisor, **Mr. Rahul Kumar Deo**, for his constant inspiration, valuable advice, and unwavering guidance throughout this investigation. I feel extremely fortunate to have had the opportunity to work alongside him and remain deeply grateful for his generosity in dedicating a significant portion of his invaluable time and expertise to this work.

My heartfelt thanks also go to all the faculty members and staff of the **Department of Computer Science & Engineering, DAVIET, Daltonganj**, for their continued support and encouragement throughout the duration of my B.Tech project.

I would like to extend my gratitude to all my friends for their assistance and cooperation during this project.

Finally, I express my deepest appreciation to my family members for their constant encouragement, patience, and understanding over the past four years.

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ABSTRACT

Templify is a modern, secure web application designed for user-centric document management. It enables authenticated users to upload, view, update, and delete various file formats—PDF, TXT, XLS, and DOC—through a RESTful API. Uploaded files are securely stored in **Cloudinary**, while metadata (such as filenames, URLs, upload timestamps, and associated user emails) is maintained in **MongoDB**, ensuring reliability and scalability.

The backend is implemented using **Node.js** and **Express**, featuring modular controllers and routes that enforce file-type validation and secure operations. The frontend, built with **React** and **TypeScript**, offers a seamless and intuitive user experience, integrating with the API to manage documents effortlessly.

This project demonstrates the integration of cloud storage, secure server-side APIs, and a modern user interface to meet real-world document management needs. It showcases the use of scalable NoSQL databases, efficient file handling via **Multer**, and environment-based configuration management with **dotenv**. The result is a robust platform capable of supporting further enhancements such as authentication layers, pagination, file previews, and role-based access.

ABBREVIATIONS

Abbreviation

API

UI

DB

JS

TS

CSS

JWT

Full Form

Application Programming
Interface

User Interface

Database

JavaScript

TypeScript

Cascading Style Sheets

JSON Web Token

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CHAPTER 1

INTRODUCTION

1.1. Project Background and Motivation

Origin Story: Templify was conceived from the observed inefficiencies and repetitive nature of manual document generation across various sectors. The idea was sparked by a personal need to automate the creation of standardized documents from raw data, which often consumed significant time and was prone to human error.

Challenges with Manual Processing: Manual document generation, particularly when dealing with large datasets or recurring needs, is fraught with challenges. These include a high propensity for errors during data transcription, considerable time consumption, and a lack of standardization, leading to inconsistencies across documents.

Vision for Automation: The initial vision for Templify was to create an automated, user-friendly solution that could seamlessly transform structured data (like CSV or Excel files) into formatted documents using predefined templates. The goal was to drastically reduce manual effort, improve accuracy, and streamline document workflows for individuals and businesses alike.

1.2. Problem Statement Addressed by Templify

The Core Problem: Businesses and individuals often struggle with repetitive document generation, requiring manual data entry into predefined formats, leading to inefficiencies and potential errors. This "last mile" of data utilization, where raw data needs to be presented in a human-readable and formalized document, remains a significant bottleneck.

Existing Solutions (and their shortcomings): While some existing solutions offer document generation, they often suffer from complexity, high costs, or a lack of user-friendly features, requiring specialized technical knowledge to operate effectively. Templify aims to provide a more intuitive, accessible, and cost-effective alternative.

1.3. Project Scope and Goals

In-Scope Functionality: Templify offers specific file type support (CSV and Excel), robust template application for document transformation, a comprehensive user history for processed files, and secure user authentication.

Out-of-Scope Functionality: For this phase, advanced template creation UI (e.g., a drag-and-drop template builder) and complex workflow automation beyond simple file-to-document transformation were not included to maintain focus and ensure timely delivery.

Key Goals:

- Develop a robust and functional web application.

- Ensure a user-friendly and intuitive interface.
- Build a maintainable and scalable codebase.
- Implement secure user authentication and data handling.

1.4. Target Audience and User Needs

Who is Templify for? Templify is designed for small businesses, administrative staff, students, and anyone who regularly needs to process structured data into formatted documents.

User Needs Addressed: Templify fulfills specific user needs, including fast and accurate document generation, secure storage of processed files, and easy access to past operations.

1.5. Document Structure

Roadmap for the Reader: This report begins with a project overview, delves into the technical stack, details the architecture, outlines challenges faced and their solutions, and concludes with future enhancements.

CHAPTER 2

Overview

2.1. Core Vision and Purpose

Recap the Vision: Templify's central idea is to serve as a tool for efficient and automated document transformation, bridging the gap between raw data and polished, formatted documents.

Value Proposition: Users gain direct benefits from Templify, including significant time savings, reduced errors, and increased productivity by automating a typically manual and error-prone process.

Design Philosophy: The application's design was guided by principles of simplicity, efficiency, and security, aiming to provide a delightful user experience through an intuitive interface and reliable functionality.

2.2. Detailed Functional Requirements

2.2.1. File Upload and Processing

- **Supported File Types:** Templify supports CSV and Excel (XLSX, XLS) file formats.
- **Drag-and-Drop / Browse:** Users can upload files through an intuitive drag-and-drop interface or by Browse their local file system.
- **File Validation:** Basic checks are performed for file type and size upon upload to ensure compatibility and prevent issues.
- **Pre-processing (if any):** Initial steps are taken after upload, such as parsing the file content, before the template application process begins.

2.2.2. Template Application and Document Transformation

- **Template Selection:** Users can choose or define a template, which for the frontend, might involve text input or selecting a predefined format.
- **Mapping Data to Template:** The conceptual process involves mapping specific data fields from the uploaded file to corresponding placeholders within the chosen template.
- **Output Format:** The transformed documents are generated in a specified output format, such as PDF or DOCX.
- **Transformation Logic:** The frontend's role is to initiate this transformation process by sending the file data and template information to the backend, and then displaying the results to the user.

2.2.3. Secure User Authentication

- **User Registration:** Users must be able to sign up for new accounts securely.

- **User Login/Logout:** Secure login and logout functionalities are provided.
- **Session Management:** User sessions are maintained securely across the application.
- **Password Management:** Basic password security measures are in place, leveraging Clerk's robust authentication capabilities.
- **Protection of User Data:** Only authenticated users can access their own data and processed files.

2.2.4. "History" Page Functionality (Review, Redownload, Delete)

- **Display Processed Files:** The "History" page lists all previously processed documents with relevant metadata (name, date, status).
- **Preview (Optional but good to mention if conceptual):** Users can view a summary or part of the processed file.
- **Redownload:** Users can re-download their generated documents from their history.
- **Delete:** Users can remove processed files from their history.
- **Search/Filter (Optional):** Ability to search or filter history.

2.2.5. User Data Privacy and Persistence

- **Data Isolation:** Each user's data and history are private and accessible only by them.
- **Session Persistence:** The user's active session and relevant application state persist across browser sessions, typically via localStorage.

2.3. Non-Functional Requirements

2.3.1. Performance and Responsiveness

- **Page Load Times:** The application is designed for fast initial load and subsequent page transitions to ensure a smooth user experience.
- **Processing Feedback:** Real-time feedback is provided to the user during file processing and API calls to indicate progress and prevent perceived delays.
- **UI Responsiveness:** The application's UI is designed to be usable and aesthetically pleasing across various screen sizes, implementing a responsive design.

2.3.2. Security and Data Integrity

- **Authentication Security:** Robust authentication is handled by Clerk, ensuring secure user access.
- **Data Transmission Security:** All communication uses HTTPS for secure data transmission.

- **Input Validation:** Frontend validation is implemented to prevent malformed data submission and improve data integrity.

2.3.3. Usability and User Experience (UX)

- **Intuitive Interface:** The interface is designed to be easy to learn and use, minimizing cognitive load for the user.
- **Clear Navigation:** A logical flow between pages ensures intuitive navigation.
- **Consistent Design:** Tailwind CSS and custom components ensure a uniform look and feel across the application.
- **Error Handling:** Errors are displayed gracefully with actionable feedback, guiding users to resolve issues.

2.3.4. Maintainability and Scalability

- **Code Structure:** The codebase is well-organized and modular, promoting easy understanding and modification.
- **Readability:** Clean, well-commented code enhances readability for developers.
- **Extensibility:** The architecture allows for easy addition of new features or modification of existing ones without major refactoring.
- **Component Reusability:** Maximizing component reuse reduces development time and improves consistency.

2.3.5. Cross-Browser Compatibility

- **Support for Modern Browsers:** The application ensures functionality across modern web browsers, including Chrome, Firefox, Edge, and Safari.

CHAPTER 3

Technology Stack (Frontend)

3.1. Overview of Frontend Architecture

Modern SPA Approach: A Single Page Application (SPA) built with React was chosen for its ability to deliver dynamic content, a rich user experience, and better performance after the initial load.

Client-Side Rendering: React renders content on the client side, allowing for interactive UIs without full page reloads.

Component-Based Paradigm: React's component model facilitates modularity and reusability, enabling the creation of independent and encapsulated UI elements.

3.2. Core Framework and Libraries

3.2.1. React with TypeScript: Foundational UI Development

Choice Justification: React was selected due to its large community, robust component-based architecture, and extensive ecosystem. TypeScript was chosen for its type safety, which significantly reduces runtime errors, improves developer experience, and enhances code maintainability, especially for larger projects.

Benefits of Type Safety: TypeScript catches errors during development, improves code understanding through explicit type definitions, and facilitates safe refactoring operations.

Component-Based Architecture: Breaking the UI into small, independent components (e.g., Button, FileUploader, HistoryItem) significantly improves development speed, reusability, and overall maintainability.

Example:

TypeScript

```
// A simple React functional component structure
```

```
interface MyButtonProps {  
  text: string;  
  onClick: () => void;  
}
```

```
const MyButton: React.FC<MyButtonProps> = ({ text, onClick }) => {  
  return (  
    <button onClick={onClick}>  
      {text}  
    </button>  
  );  
};
```

3.2.2. Vite: Accelerated Development and Optimized Builds

Choice Justification: Vite was chosen over alternatives like Webpack or Create React App for its exceptional speed, modern approach to frontend tooling, and built-in optimizations.

Fast Refresh and Hot Module Replacement (HMR): These features drastically reduce development iteration times by instantly reflecting code changes in the browser without requiring a full page reload, leading to a much more efficient development workflow.

Production Build Optimization Strategies: Vite automatically optimizes for production builds through strategies like code splitting, minification, and tree-shaking, resulting in smaller bundle sizes and faster load times for end-users.

3.3. Styling and UI Development

3.3.1. Tailwind CSS: Utility-First Approach for Rapid UI

Choice Justification: Tailwind CSS was selected for its utility-first approach, which enables rapid prototyping, eliminates the need for complex, highly specific CSS files, and promotes a highly customizable and consistent design system.

Utility-First Concept: "Utility-first" means applying pre-defined, single-purpose CSS classes directly in the HTML markup. This contrasts with traditional CSS or CSS-in-JS by allowing for highly granular styling without writing custom CSS rules for every element.

Customization and Theme Management: The `tailwind.config.js` file allows for easy customization of design tokens such as colors, fonts, spacing, and breakpoints, ensuring brand consistency and quick theme adjustments across the application.

Integration with PostCSS: Tailwind leverages PostCSS for processing, which allows for powerful transformations of CSS with JavaScript plugins.

3.3.2. Custom UI Components (Inspired by shadcn/ui)

Design Principles: The philosophy behind creating custom components was rooted in composability, accessibility, and a headless UI approach. This means components are designed to be easily combined, are usable by individuals with disabilities, and provide the core logic without opinionated styling.

Component Reusability: Components housed in the `ui/` directory (e.g., Button, Card, Input) provide a consistent visual language across the application and significantly reduce redundant code, contributing to a more maintainable codebase.

Accessibility Considerations: Efforts were made to ensure components are accessible, incorporating proper ARIA attributes and supporting keyboard navigation to accommodate users with various needs.

Example:

TypeScript

```
// A simple example of a custom button component
import React from 'react';
import { cn } from '@lib/utls'; // A utility to combine Tailwind classes

interface ButtonProps extends React.ButtonHTMLAttributes<HTMLButtonElement> {
```

```

    variant?: 'primary' | 'secondary' | 'ghost';
  }

const Button: React.FC<ButtonProps> = ({ className, variant = 'primary', children, ...props }) => {
  return (
    <button
      className={cn(
        'px-4 py-2 rounded-md font-semibold transition-colors duration-200',
        variant === 'primary' && 'bg-blue-600 text-white hover:bg-blue-700',
        variant === 'secondary' && 'bg-gray-200 text-gray-800 hover:bg-gray-300',
        variant === 'ghost' && 'bg-transparent text-blue-600 hover:bg-blue-100',
        className
      )}
      {...props}
    >
      {children}
    </button>
  );
};

export { Button };

```

3.3.3. Global CSS for Base Styles

Purpose: While Tailwind is used for most styling, global CSS handles fundamental resets (e.g., browser default styles), root variables for theming, or very broad application-wide styles that are not specific to individual components.

3.4. State Management and Data Persistence

3.4.1. React Context API: Global State Management

Choice Justification: The React Context API was chosen for global state management because it is sufficient for medium-sized applications, effectively avoids "prop drilling" (passing props down multiple levels of the component tree), is built into React, and has less overhead compared to more complex state management libraries like Redux for the application's current needs.

FileContext for File-Related Data: This context manages all state related to file uploads, the processing status of files, and the resulting transformed file data.

HistoryContext for User Processing History: This context's role is to manage the list of previously processed files and their associated metadata, enabling the "History" page functionality.

How it Works: The Provider/Consumer pattern of the Context API is utilized, where a Provider component wraps a part of the component tree, making the context's value available to all consuming components within that tree.

3.4.2. localStorage Integration

Purpose: localStorage is integrated into the application to persist certain aspects of the application state, specifically within FileContext, across different browser sessions. This ensures that user's recent interactions or loaded files are not lost if they close and reopen the browser.

Implementation: useEffect hooks are used within the context provider. One useEffect hook triggers whenever FileContext's state changes, serializing the state (e.g., using JSON.stringify) and saving it to localStorage under a specific key. Another useEffect hook runs once on component mount or application load, retrieving the stored state from localStorage (using JSON.parse) and initializing the FileContext with it.

Benefits: This integration significantly improves the user experience by remembering the user's last interaction or loaded file, enhancing convenience and reducing repetitive actions.

3.5. Routing and Navigation

3.5.1. React Router: Client-Side Routing Implementation

Choice Justification: React Router was chosen as the client-side routing library due to its status as an industry standard, its robustness, and its declarative approach to defining routes.

Defining Routes: Routes are clearly defined for different sections of the application, such as / (Home), /history, /sign-in, and /sign-up.

Seamless Navigation: React Router enables fast, single-page application navigation without requiring full page reloads, providing a fluid user experience.

Programmatic Navigation: The useNavigate hook is utilized for programmatic navigation, allowing the application to redirect users based on certain actions or conditions (e.g., after a successful login).

Route Protection (if applicable): Routes are protected to ensure that unauthenticated users are redirected from pages requiring authentication, such as the /history page.

3.6. Authentication System

3.6.1. Clerk: Secure User Authentication Management

Choice Justification: Clerk was chosen for authentication management due to its ability to handle authentication complexities, its inherent security features, ease of integration, and provision of pre-built components and hooks.

Sign-Up, Sign-In, and Session Management: Clerk efficiently manages user registration, login processes, and maintains secure user sessions across the application.

Integration with Frontend Components and API: Clerk seamlessly integrates with React components via its dedicated hooks, and it can provide authentication tokens necessary for secure backend API calls.

3.7. User Feedback and Notifications

3.7.1. react-hot-toast: Non-Intrusive Notifications

Purpose: react-hot-toast is used to provide ephemeral, non-blocking feedback to the user, ensuring that notifications are informative without disrupting the user's workflow.

Use Cases: It is utilized for various notifications, such as "File uploaded successfully!", "Processing...", and displaying error messages like "Error: Invalid input".

Customization: The library offers the ability to customize toast appearance and types (e.g., success, error, loading) to match the application's design and provide clear visual cues.

3.8. Code Quality and Development Tools

3.8.1. ESLint: Static Code Analysis and Style Enforcement

Purpose: ESLint is used to perform static code analysis, identify potential problems, and ensure that the codebase adheres to predefined coding standards and best practices.

Configuration: The `.eslintrc.cjs` setup integrates with React, TypeScript, and Prettier configurations to enforce consistent code style and identify common errors.

Benefits: ESLint helps catch errors early in the development cycle, promotes consistent coding practices across the team (if applicable), and ultimately reduces the likelihood of bugs.

3.8.2. Prettier: Code Formatting for Consistency

Purpose: Prettier is an opinionated code formatter that automatically formats code, ensuring a consistent style across the entire codebase regardless of who wrote the code.

Integration: It works seamlessly with ESLint and most Integrated Development Environments (IDEs), often formatting code on save.

Benefits: Prettier reduces cognitive load for developers by removing style debates, makes code easier to read, and streamlines code reviews by focusing on logic rather than formatting.

CHAPTER 4

Frontend Architecture and File Structure

4. Frontend Architecture and File Structure

This section is about how you organized your code, justifying your choices for a clean and scalable project.

4.1. Principles of Modular Design

Separation of Concerns: Different parts of the application are kept distinct, such as UI components separate from business logic, and contexts separate from pages. This ensures that each module has a single responsibility.

Modularity: The application is broken down into smaller, independent, and reusable modules. This approach enhances understandability and manageability.

Scalability: This structure allows for easy addition of new features without disrupting existing ones, promoting future growth and expansion of the application.

Maintainability: The modular design makes the codebase easier to understand, debug, and update, significantly reducing the effort required for long-term maintenance.

4.2. Detailed Directory Breakdown

public/:

- **Purpose:** Contains static assets that are served directly by the web server without being processed by Vite's build pipeline.
- **Contents:** vite.svg (example), index.html (the root HTML file).

src/:

- **Purpose:** The main application source code.

assets/:

- **Purpose:** Stores images, icons, fonts, and other static media used throughout the application.
- **icons/:** A specific subdirectory for SVG or other icon files, promoting organization of visual assets.

components/:

- **Purpose:** Houses all reusable UI components.
- **ui/ Base UI elements:** These are generic, highly reusable components (e.g., Button.tsx,

Card.tsx, Input.tsx) often built from primitive HTML elements. They can follow headless UI principles and form the foundational building blocks of the design system, providing a consistent look and feel.

- **Application-Specific Components:** Components that are unique to Templify's features but are still reusable within different pages (e.g., FileUploader.tsx for handling file uploads, AppLayout.tsx for common layout structures like navigation bars and footers).

context/:

- **Purpose:** Contains React Context API providers and consumers for global state management.
- **FileContext.tsx:** Manages state related to file uploads, processing status, and the results of file transformations.
- **HistoryContext.tsx:** Manages the state of the user's processed file history, enabling display and interaction with past operations.
- **Justification:** Centralizes state that needs to be accessed by multiple components across the application, effectively avoiding "prop drilling" for widely used data.

hooks/:

- **Purpose:** Custom React Hooks that encapsulate reusable logic. These hooks allow for sharing stateful logic across components without duplicating code.
- **useFileContext.ts:** A custom hook to easily consume FileContext, simplifying access to file-related global state.
- **useFileHandling.ts:** Encapsulates logic related to file selection, validation, and initial processing steps, abstracting these concerns from components.
- **useTemplateEditor.ts:** (Crucial for refactoring). This hook plays a vital role in abstracting business logic from Home.tsx, keeping the main page component clean and focused on rendering. It would contain logic for template input, data mapping, and initiating backend processing.
- **Justification:** Promotes logic reuse across components, keeps components clean and focused on rendering their UI, and improves testability by isolating logic.

lib/:

- **Purpose:** General utility functions, API clients, and helper modules that don't directly relate to UI components or state management. This directory often contains generic, non-UI specific functionalities.
- **gemini.ts:** If utilizing the Gemini API, this file would handle its integration, such as managing AI-driven template suggestions or content generation.

- **utils.ts:** Miscellaneous helper functions, such as date formatting, string manipulation, or other small, reusable helper functions that don't fit into more specific categories.

pages/:

- **Purpose:** Top-level components that represent full-page views of the application. These pages typically compose multiple smaller components/ to form a complete UI.
- **Home.tsx:** The main page for file upload and template application, serving as the primary interface for users to interact with Templify's core functionality.
- **History.tsx:** Displays the user's processing history, allowing them to review, redownload, or delete previously processed documents.
- **SignIn.tsx:** The user sign-in interface.
- **SignUp.tsx:** The user registration interface.
- **Justification:** Provides a clear mapping of application routes to specific views, making it easy to understand the application's navigation structure.

App.tsx:

- **Purpose:** The main application component responsible for setting up client-side routing (using React Router), configuring context providers for global state, and defining the overall layout structure of the application.

main.tsx:

- **Purpose:** The entry point of the React application, responsible for rendering the App component into the DOM, initiating the entire frontend application.

General Justification for src/ Structure: This hierarchical structure promotes a clear mental model of the application, making it easier for developers to navigate the codebase and understand where specific functionalities reside. It significantly supports scaling the application by allowing new features to be added in a consistent and organized manner without negatively impacting existing code.

4.3. Configuration Files

.eslintrc.cjs:

- **Purpose:** Configuration file for ESLint, defining rules for code style, identifying potential errors, and enforcing best practices within the JavaScript/TypeScript codebase.
- **Impact:** Ensures code consistency and quality across the project, helping to catch issues early in development.

tailwind.config.js:

- **Purpose:** Configuration file for Tailwind CSS, allowing extensive customization of themes, colors, fonts, spacing, and the addition of custom utility classes.

- **Impact:** Ensures a consistent design system and facilitates rapid styling by centralizing design token definitions.

tsconfig.json:

- **Purpose:** TypeScript compiler configuration file, defining how TypeScript files are transpiled to JavaScript. It specifies compiler options, root directories, and type definitions.
- **Impact:** Enables type checking, ensures strong typing throughout the project, and guarantees correct compilation for different environments.

vite.config.ts:

- **Purpose:** Configuration file for Vite, defining build settings, development server options, and plugins. This includes configurations for handling different file types, proxy settings, and build optimizations.
- **Impact:** Optimizes the build process for fast development and efficient production bundles, and configures the development server for hot module replacement and other features.

CHAPTER 5

Development Challenges and Solutions

5.1. Approach to Problem Solving and Iterative Development

Methodology: Our approach to tackling problems involved identifying the root cause, thoroughly researching potential solutions, implementing changes iteratively, and conducting rigorous testing. This iterative cycle allowed for continuous refinement and adaptation.

Importance of Refactoring: Refactoring was not merely seen as "cleaning up" code, but as a crucial part of improving code quality, maintainability, and overall system health. It proactively addressed technical debt and prepared the codebase for future enhancements.

5.2. Challenge 1: Syncing Frontend with Backend API Changes

Problem Description:

Initial State: The UI was initially built based on anticipated or mock data structures, which is common in parallel frontend and backend development.

Discrepancy: As the backend API evolved, discrepancies emerged, such as field name mismatches (e.g., `id` versus `_id` for unique identifiers) and differences in download logic (frontend attempting to generate URLs versus the backend providing direct download URLs).

Impact: These inconsistencies led to broken features, incorrect data display on the UI, and inefficient client-side logic that was attempting to compensate for expected but absent backend behaviors.

Solution Implemented:

HistoryContext Update: The `HistoryContext` was modified to correctly parse and map the incoming API response fields to the expected frontend data structure. This involved creating data transformation logic within the context to ensure that `History.tsx` received data in a consistent and usable format.

History.tsx Component Adjustment: The component's rendering logic was updated to seamlessly use the new, correctly mapped data from the `HistoryContext`. This involved updating how data was accessed and displayed to reflect the transformed structure.

handleDownload Refactoring: The logic for `handleDownload` was refactored. Instead of client-side URL generation, the application now directly utilizes the `downloadURL` provided by the backend API, simplifying the download process and ensuring accuracy.

Example:

TypeScript

```
// In HistoryContext.tsx (Simplified illustrative example)
// Before (mock data approach)
// const mappedHistoryItem = {
```

```
// id: apitem.id, // Assumed 'id' from API
// downloadUrl: `/files/${apitem.id}/download`, // Frontend constructs URL
// // ... other fields
// };

// After (API driven approach)
const mappedHistoryItem = {
  id: apitem._id, // Corrected to '_id' from API
  downloadURL: apitem.downloadURL, // Directly use downloadURL from API
  fileName: apitem.fileName,
  processedDate: apitem.createdAt,
  status: apitem.status,
  // ... other fields
};

// In History.tsx (Simplified illustrative example)
// Before (accessing derived URL)
// <a href={item.downloadUrl}>Download</a>
```

```
// After (accessing API-provided URL)
<a href={item.downloadURL} target="_blank" rel="noopener noreferrer">Download</a>
```

Lessons Learned: This challenge underscored the critical importance of clear API contracts between frontend and backend teams, continuous communication throughout the development lifecycle, and the need for flexible data mapping on the frontend to gracefully handle evolving API structures.

5.3. Challenge 2: Improving State Persistence and User Experience

Problem Description:

Initial State: A significant user experience issue arose when navigating away from Home.tsx to History.tsx and then back to Home.tsx. Any loaded files or text input in the Home.tsx component would be cleared, forcing the user to re-upload or re-enter data.

Impact: This lack of state persistence led to a frustrating user experience, potential loss of work, and reduced overall productivity, as users had to repeat actions unnecessarily.

Solution Implemented:

Enhancing FileContext: localStorage was integrated directly into the FileContext. This allowed critical file-related state to be saved and retrieved persistently across browser sessions.

useEffect Hooks for Persistence:

- **Saving State:** A useEffect hook was implemented to trigger whenever the FileContext's state changed. Inside this hook, the relevant state (e.g., the uploaded file object, template text) was serialized using JSON.stringify and saved to localStorage under a specific key.
- **Loading State:** Another useEffect hook was implemented to run once on component mount

or application load. This hook retrieved the stored state from localStorage (using JSON.parse) and used it to initialize the FileContext, effectively restoring the user's previous interaction state.

Example:

TypeScript

```
// In FileContext.tsx (Simplified)
import React, { createContext, useState, useEffect, useContext } from 'react';

interface FileState {
  uploadedFile: File | null;
  templateInput: string;
  // ... other relevant state
}

const initialFileState: FileState = {
  uploadedFile: null,
  templateInput: "",
};

const FileContext = createContext<{
  fileState: FileState;
  setFileState: React.Dispatch<React.SetStateAction<FileState>>;
} | undefined>(undefined);

export const FileProvider: React.FC<{ children: React.ReactNode }> = ({ children }) => {
  const [fileState, setFileState] = useState<FileState>(() => {
    // Load state from localStorage on initial render
    const storedState = localStorage.getItem('fileContextState');
    if (storedState) {
      try {
        // Attempt to parse stored state, handle potential errors
        return JSON.parse(storedState);
      } catch (e) {
        console.error("Failed to parse fileContextState from localStorage", e);
        return initialFileState;
      }
    }
  });
  return initialFileState;
});

// Save state to localStorage whenever it changes
useEffect(() => {
```

```

    localStorage.setItem('fileContextState', JSON.stringify(fileState));
  }, [fileState]); // Dependency array: run effect when fileState changes

  return (
    <FileContext.Provider value={{ fileState, setFileState }}>
      {children}
    </FileContext.Provider>
  );
};

export const useFileContext = () => {
  const context = useContext(FileContext);
  if (context === undefined) {
    throw new Error('useFileContext must be used within a FileProvider');
  }
  return context;
};

```

Impact: This solution resulted in a seamless user experience, significantly improved retention of user progress within the application, and overall enhanced usability by reducing the need for repeated data entry.

5.4. Challenge 3: Refactoring for Maintainability and Scalability

Problem Description:

The "God Component": The Home.tsx component initially grew too large and complex. It contained a mix of UI rendering logic and intricate business logic, including file handling, template application, API calls, and local state management for various inputs (e.g., template text, file preview data).

Impact: This "God Component" anti-pattern made the Home.tsx file extremely difficult to understand, test, and debug. Any modification carried a high risk of introducing new bugs due to tightly coupled functionalities. Furthermore, parts of the logic were impossible to reuse in other components.

Solution Implemented:

Creating useTemplateEditor Custom Hook:

- **Purpose:** A new custom React Hook, useTemplateEditor, was created specifically to abstract all business logic related to template editing and file processing away from the Home.tsx component.
- **Content:** All local state (e.g., for text input, selected template, processing status), event handlers (e.g., handleFileUpload, handleSubmit for processing), side effects (e.g., API calls to the backend, input validation), and derived state were moved into this hook.
- **Return Values:** The useTemplateEditor hook returns only the necessary values and functions that Home.tsx needs to render its UI and interact with the logic (e.g., fileData, templateText, isLoading, onUpload, onSubmit, processingError). **Home.tsx**

Transformation: As a result of this refactoring, Home.tsx was transformed into a pure, presentational component. Its primary responsibility became rendering UI elements and passing props to its child components. All its complex logic was now consumed through the useTemplateEditor hook, making it significantly cleaner and more readable.

Example:

TypeScript

```
// In hooks/useTemplateEditor.ts (Conceptual)
import { useState, useCallback } => 'react';
import { toast } => 'react-hot-toast'; // Assuming react-hot-toast for notifications

function useTemplateEditor() {
  const [uploadedFile, setUploadedFile] = useState<File | null>(null);
  const [templateText, setTemplateText] = useState<string>("");
  const [isProcessing, setIsProcessing] = useState<boolean>(false);
  const [processResult, setProcessResult] = useState<any>(null); // Placeholder for processed data
  const [error, setError] = useState<string | null>(null);

  const handleFileUpload = useCallback((file: File) => {
    // Logic for file validation and setting uploaded file
    setUploadedFile(file);
    toast.success(`File "${file.name}" uploaded.`);
  }, []);

  const handleSubmit = useCallback(async () => {
    if (!uploadedFile || !templateText) {
      toast.error("Please upload a file and enter template text.");
      return;
    }

    setIsProcessing(true);
    setError(null);
    const loadingToastId = toast.loading("Processing document...");

    try {
      // Simulate API call to backend for document transformation
      // In a real app, this would involve fetch or axios
      const response = await new Promise((resolve) => setTimeout(() => {
        // Simulate success or failure
        if (Math.random() > 0.1) {
          resolve({ success: true, transformedData: { /* ... */ } });
        } else {
          throw new Error("Simulated processing error.");
        }
      }, 1000));
    } catch (error) {
      // Error handling logic
    }
  }, [uploadedFile, templateText]);
}
```

```

    }
    }, 2000));

    if ((response as any).success) {
        setProcessResult((response as any).transformedData);
        toast.success("Document processed successfully!", { id: loadingToastId });
    } else {
        throw new Error("Processing failed.");
    }
} catch (err: any) {
    setError(err.message || "An error occurred during processing.");
    toast.error(err.message || "Failed to process document.", { id: loadingToastId });
} finally {
    setIsProcessing(false);
}
}, [uploadedFile, templateText]);

return {
    uploadedFile,
    setUploadedFile,
    templateText,
    setTemplateText,
    isProcessing,
    processResult,
    error,
    handleFileUpload,
    handleSubmit,
};
}

export { useTemplateEditor };

// In pages/Home.tsx (Simplified)
import React from 'react';
import { useTemplateEditor } from '../hooks/useTemplateEditor';
import { Button } from '../components/ui/Button'; // Assuming your custom Button component
import { FileUploader } from '../components/FileUploader'; // Assuming your FileUploader component

const Home: React.FC = () => {
    const {
        uploadedFile,
        setUploadedFile, // If Home needs to allow clearing file
        templateText,

```

```

    setTemplateText,
    isProcessing,
    processResult,
    error,
    handleFileUpload,
    handleSubmit,
  } = useTemplateEditor();

  return (
    <div className="home-page">
      <h1>Templify Document Processor</h1>
      <FileUploader onFileUpload={handleFileUpload} />
      {uploadedFile && (
        <p>File selected: {uploadedFile.name}</p>
      )}
      <textarea
        placeholder="Enter your template text here..."
        value={templateText}
        onChange={(e) => setTemplateText(e.target.value)}
        rows={10}
        className="w-full p-2 border rounded"
      ></textarea>
      <Button onClick={handleSubmit} disabled={isProcessing}>
        {isProcessing ? 'Processing...' : 'Process Document'}
      </Button>
      {processResult && (
        <div className="mt-4 p-4 border rounded bg-green-50">
          <h2 className="text-lg font-semibold">Processed Result:</h2>
          <pre>JSON.stringify(processResult, null, 2)</pre>
        </div>
      )}
      {error && (
        <div className="mt-4 p-4 border rounded bg-red-50 text-red-700">
          Error: {error}
        </div>
      )}
    </div>
  );
};

```

export default Home;

Benefits: This refactoring dramatically improved code readability, testability, and maintainability. It also made the business logic highly reusable, as useTemplateEditor could potentially be integrated into other

components that require similar processing capabilities, greatly enhancing the scalability of the application.

5.5. Challenge 4: Handling Authentication Errors Gracefully

Problem Description:

Clerk's Default Behavior: Initially, the pre-built Clerk `<SignIn>` and `<SignUp>` components, while convenient, had a default behavior of redirecting users to a Clerk-hosted page whenever an authentication error occurred (e.g., invalid credentials, email already in use).

Impact: This redirection resulted in a disjointed and jarring user experience. Users were taken away from the application's context, which could lead to confusion and a perception of a broken flow.

Solution Implemented:

Custom Forms with Clerk Hooks: The default Clerk components were replaced with entirely custom React forms. This provided full control over the UI and user interaction during authentication.

`useSignIn` and `useSignUp` Hooks: Clerk's specific hooks (`useSignIn`, `useSignUp`) were utilized. These hooks provide fine-grained control over the authentication process, allowing developers to manage submission, loading states, and error handling directly within the application's components.

`try...catch` Blocks: The authentication submission logic within the custom forms was wrapped inside `try...catch` blocks. This allowed for the interception of errors thrown by Clerk's API calls (e.g., `signIn.authenticateWithPassword`).

Error Extraction: The error objects returned by Clerk were parsed to extract user-friendly error messages (e.g., "Invalid credentials", "Email already exists"). Clerk's error structure typically provides detailed error codes and messages.

`react-hot-toast` for Feedback: The extracted, user-friendly error messages were then displayed using `react-hot-toast`. This provided immediate, non-intrusive, in-app feedback to the user without any jarring redirections.

Example:

TypeScript

```
// In pages/SignIn.tsx (Simplified)
import React, { useState } from 'react';
import { useSignIn } from '@clerk/clerk-react';
import { useNavigate } from 'react-router-dom';
import { toast } from 'react-hot-toast';
import { Button } from '../components/ui/Button'; // Your custom Button
import { Input } from '../components/ui/Input'; // Your custom Input

const SignIn: React.FC = () => {
  const { signIn, isLoading, setActive } = useSignIn();
  const navigate = useNavigate();
```

```

const [identifier, setIdentifier] = useState("");
const [password, setPassword] = useState("");
const [isLoading, setIsLoading] = useState(false);

const handleSubmit = async (e: React.FormEvent) => {
  e.preventDefault();
  if (!isLoading) {
    return; // Clerk is not yet loaded
  }

  setIsLoading(true);
  let loadingToastId: string | undefined;
  try {
    loadingToastId = toast.loading("Signing in...");
    const result = await signIn.authenticateWithPassword({
      identifier,
      password,
    });

    if (result.status === 'complete') {
      await setActive({ session: result.createdSessionId });
      toast.success("Signed in successfully!", { id: loadingToastId });
      navigate('/home'); // Redirect on success
    } else {
      // This path might be for MFA or other intermediate steps, handled by Clerk's flow
      // For simple username/password, usually 'complete' or an error is returned
      toast.error("Authentication incomplete. Please check for further steps.", { id: loadingToastId });
    }
  } catch (err: any) {
    console.error('Clerk sign-in error:', err);
    // Extract user-friendly message from Clerk's error object
    // err.errors is an array, take the first error's longMessage
    const errorMessage = err.errors?.[0]?.longMessage || "An unknown error occurred.";
    toast.error(errorMessage, { id: loadingToastId });
  } finally {
    setIsLoading(false);
  }
};

return (
  <div className="flex justify-center items-center h-screen bg-gray-100">
    <form onSubmit={handleSubmit} className="bg-white p-8 rounded-lg shadow-md w-96">
      <h2 className="text-2xl font-bold mb-6 text-center">Sign In</h2>

```

```

    <div className="mb-4">
      <label htmlFor="identifier" className="block text-gray-700 text-sm font-bold mb-2">
        Email or Username
      </label>
      <Input
        id="identifier"
        type="text"
        value={identifier}
        onChange={(e) => setIdentifier(e.target.value)}
        required
        placeholder="Enter your email or username"
      />
    </div>
    <div className="mb-6">
      <label htmlFor="password" className="block text-gray-700 text-sm font-bold mb-2">
        Password
      </label>
      <Input
        id="password"
        type="password"
        value={password}
        onChange={(e) => setPassword(e.target.value)}
        required
        placeholder="Enter your password"
      />
    </div>
    <Button type="submit" disabled={!isLoading || isLoading} className="w-full">
      {isLoading ? 'Signing In...' : 'Sign In'}
    </Button>
    <p className="mt-4 text-center text-sm text-gray-600">
      Don't have an account?{' '}
      <button type="button" onClick={() => navigate('/sign-up')} className="text-blue-600
hover:underline">
        Sign Up
      </button>
    </p>
  </form>
</div>
);
};

```

export default SignIn;

Outcome: This approach resulted in a much smoother, integrated, and user-friendly authentication

experience by keeping the user within the application context and providing immediate, clear feedback for any authentication issues.

5.6. Challenge 5: Optimizing Data Fetching and Performance

Problem Description:

Redundant API Calls: The application was initially prone to refetching history data unnecessarily. This could happen on every component re-render or even during routine navigation, leading to inefficient network usage.

Lack of User Feedback: During API calls (e.g., file uploads, history fetching), there were no visual indicators, leading to a perceived slow application performance. Users were left guessing if an action was being processed.

Multiple Loading Indicators: Due to uncontrolled fetch calls and lack of proper management, loading toasts would sometimes appear and disappear erratically, creating a confusing user experience.

Solution Implemented:

Conditional Fetching with useLocation: React Router's useLocation hook was strategically utilized. This hook provides access to the current URL location, enabling a mechanism to ensure that the fetchHistory API call was only initiated when the user was specifically on the /history route. This prevented unnecessary data fetches when navigating to other parts of the application.

Preventing Duplicate Fetches with useRef: A useRef flag (e.g., hasFetched.current = true) was implemented. This flag, being mutable and persistent across renders without causing re-renders itself, ensured that the fetchHistory operation would run only once per component mount (or specific condition), effectively preventing redundant calls on subsequent re-renders within the same page visit.

Loading Toasts for User Feedback:

- **Initiation:** A loading toast (toast.loading()) was displayed immediately before any asynchronous operation began (e.g., file upload, history fetch, sign-in attempt). This provided instant visual feedback to the user that an action was in progress.
- **Resolution:** Upon the completion or failure of the asynchronous operation, the loading toast was updated to a success (toast.success()) or error (toast.error()) message. This provided clear, consistent, and actionable feedback to the user about the outcome of their action.

Example:

TypeScript

```
// In pages/History.tsx or a custom hook for history management
import React, { useEffect, useRef, useState } from 'react';
import { useLocation } from 'react-router-dom';
import { toast } from 'react-hot-toast';
// Assuming fetchHistory is an async function that fetches history data
import { fetchHistoryData } from '../lib/apiClient'; // Example API client
```

```

interface HistoryItem {
  id: string;
  fileName: string;
  processedDate: string;
  status: string;
  downloadURL: string;
}

const History: React.FC = () => {
  const location = useLocation();
  const hasFetched = useRef(false); // Flag to prevent multiple fetches
  const [historyItems, setHistoryItems] = useState<HistoryItem[]>([]);
  const [isLoadingHistory, setIsLoadingHistory] = useState(false);
  const [error, setError] = useState<string | null>(null);

  useEffect(() => {
    // Only fetch history if on the /history route and hasn't fetched yet
    if (location.pathname === '/history' && !hasFetched.current) {
      const loadingToastId = toast.loading("Fetching history...");
      setIsLoadingHistory(true);
      setError(null);

      fetchHistoryData()
        .then((data: HistoryItem[]) => {
          setHistoryItems(data);
          toast.success("History loaded!", { id: loadingToastId });
          hasFetched.current = true; // Mark as fetched
        })
        .catch((err) => {
          setError(err.message || "Failed to load history.");
          toast.error("Failed to load history.", { id: loadingToastId });
        })
        .finally(() => {
          setIsLoadingHistory(false);
        });
    }
  }, [location.pathname]); // Re-run effect if pathname changes

  // Other component logic for rendering history items, redownload, delete etc.
  return (
    <div className="history-page p-4">
      <h1 className="text-2xl font-bold mb-4">Your Processing History</h1>

```



```

    {isLoadingHistory && <p>Loading history...</p>}
    {error && <p className="text-red-500">Error: {error}</p>}
    {!isLoadingHistory && historyItems.length === 0 && !error && (
      <p>No processed files found yet. Start by uploading a file on the Home page!</p>
    )}
    <ul className="space-y-4">
      {historyItems.map((item) => (
        <li key={item.id} className="p-4 border rounded-md shadow-sm flex justify-between items-
center">
          <div>
            <p className="font-semibold">{item.fileName}</p>
            <p className="text-sm text-gray-500">Processed on: {new
Date(item.processedDate).toLocaleDateString()}</p>
            <p className="text-sm text-gray-500">Status: {item.status}</p>
          </div>
          <div className="flex space-x-2">
            <a href={item.downloadURL} target="_blank" rel="noopener noreferrer"
className="text-blue-600 hover:underline">
              Redownload
            </a>
            <button
              onClick={() => {/* handle delete logic */ toast.success('Deleted!');}}
              className="text-red-600 hover:underline"
            >
              Delete
            </button>
          </div>
        </li>
      ))}
    </ul>
  </div>
);
};

```

export default History;

Results: These optimizations significantly improved application performance by reducing unnecessary network requests and API calls. More importantly, they led to a dramatically better user experience due to clear, consistent, and non-intrusive feedback during asynchronous operations, making the application feel more responsive and reliable.

CHAPTER 6

Frontend Development Process and Methodologies

6.1. Agile Principles Applied

Iterative Development: The project followed an iterative development approach, breaking down the project into smaller, manageable iterations or sprints. Each iteration focused on delivering a functional piece of the application, allowing for continuous progress and early feedback.

Flexibility and Adaptability: We embraced flexibility and adaptability, responding to changing requirements or unforeseen challenges (e.g., API changes, refactoring needs) by adjusting plans and priorities as necessary. This agile mindset prevented rigid adherence to initial plans when new information emerged.

Continuous Improvement: We continuously reviewed and improved our code and processes. Regular code reviews, retrospectives (if applicable), and refactoring efforts were integral to enhancing the quality of the codebase and the efficiency of the development workflow.

6.2. Version Control with Git and GitHub

Branching Strategy: A standard Git branching model was employed, typically involving a main branch for production-ready code, a develop branch for ongoing development, and feature branches for new functionalities. This ensured stable releases while allowing for parallel development.

Commit Hygiene: Emphasis was placed on writing descriptive commit messages that clearly communicated the purpose and scope of each change, making the commit history understandable and aiding in debugging.

Pull Requests (PRs): If working in a team, code reviews were conducted via Pull Requests (PRs), ensuring code quality, knowledge sharing, and adherence to coding standards before merging into develop or main.

Collaboration: Git and GitHub facilitated collaborative development by providing a centralized platform for code sharing, version tracking, and issue management, even for solo projects for better organization.

6.3. Testing Strategy (Unit, Integration, E2E - if applicable)

Manual Testing: Extensive manual testing was performed during the development of each feature to ensure it met functional requirements and provided a good user experience.

Unit Tests (If implemented): If unit tests were written (e.g., using Jest/React Testing Library), they focused on isolated units of code, such as individual component rendering, custom hook logic, and utility functions, ensuring their correct behavior.

Integration Tests (If implemented): Integration tests focused on verifying the interactions between different parts of the frontend, such as how components interact with context providers or how UI actions

trigger API calls.

End-to-End (E2E) Tests (Less likely for a solo project, but mention if considered): While less common for solo projects due to time constraints, consideration was given to automated E2E tests (e.g., using Cypress, Playwright) to test full user flows from a user's perspective.

Importance of Testing: Testing was crucial for ensuring the reliability of the application, preventing regressions, and boosting confidence in making changes to the codebase.

6.4. Deployment Pipeline (briefly, if applicable)

Hosting Platform: The frontend application is hosted on a modern platform like Vercel or Netlify, known for their seamless integration with Git repositories.

Deployment Process: Changes are deployed automatically upon pushes to the main branch, enabling continuous deployment and rapid iteration.

CI/CD (Continuous Integration/Continuous Deployment): Basic CI/CD practices were adopted, such as automated builds and linting checks triggered by Git commits, ensuring code quality before deployment.

CHAPTER 7

Conclusion and Future Enhancement

7.1. Summary of Frontend Development Achievements

Project Success: The successful delivery of Templify's frontend marks a significant achievement, providing a robust and intuitive platform for automated document generation.

Key Learnings: Significant technical and project management lessons were learned, including the importance of robust state management, efficient API integration, and modular component design.

Quality of Deliverable: The current application is robust, performant, and user-friendly, demonstrating a high quality of deliverable that meets the initial project goals.

7.2. Contributions to Templify's Overall Success

User Engagement: The frontend's intuitive design and seamless functionality directly contribute to user satisfaction and engagement, making the complex process of document transformation accessible.

Core Functionality Enablement: The frontend provides the essential interface through which users interact with and leverage the backend's powerful capabilities for document processing.

Scalability Foundation: The chosen technologies and architectural patterns provide a strong, maintainable, and scalable foundation, allowing for future growth and the seamless integration of new features.

7.3. Potential Future Enhancements for the Frontend

Advanced Template Management UI:

- **Problem:** Currently, templates might be simple text inputs or predefined formats, limiting user flexibility.
- **Enhancement:** Develop a rich visual editor for creating and managing templates directly within the application, allowing users to define fields, customize layout, and apply styling in an intuitive, drag-and-drop manner.

Real-time Collaboration Features:

- **Problem:** The current application is designed for a single-user experience.
- **Enhancement:** Implement features that allow multiple users to collaborate on template creation or document processing in real-time, enhancing teamwork and productivity.

Internationalization and Localization (i18n & l10n):

- **Problem:** The application currently supports only a single language.
- **Enhancement:** Introduce support for multiple languages to cater to a global user base, making Templify accessible to a wider audience.

Performance Monitoring and Optimization (further):

- **Problem:** While basic optimizations were done, continuous monitoring is crucial for long-term performance.
- **Enhancement:** Integrate tools for real-time performance monitoring (e.g., Lighthouse CI, Web Vitals monitoring) and implement more advanced lazy loading or code-splitting strategies to ensure optimal performance as the application scales.

Progressive Web App (PWA) Capabilities:

- **Problem:** The application is not currently installable or accessible offline.
- **Enhancement:** Enable PWA capabilities, allowing users to install the application to their home screen, access basic functionalities offline, and receive push notifications for a more native-like experience.

CHAPTER 8

Backend

8.1. Technologies Used

a. Node.js

Purpose: JavaScript runtime for server-side development.

Why: Non-blocking, event-driven architecture ideal for I/O-heavy applications like file uploads.

b. Express.js

Purpose: Web framework for Node.js to build RESTful APIs.

Why: Simplifies routing, middleware, and request/response handling.

c. MongoDB (with Mongoose)

Purpose: NoSQL database for storing user and file metadata.

Why: Flexible schema, easy to scale, and integrates well with Node.js via Mongoose ODM.

d. Cloudinary

Purpose: Cloud storage for user-uploaded files.

Why: Handles file storage, access, and delivery efficiently; supports direct file URLs.

e. Multer

Purpose: Middleware for handling multipart/form-data (file uploads) in Express.

Why: Simplifies file upload handling and integrates with Express routes.

f. dotenv

Purpose: Loads environment variables from a .env file.

Why: Keeps sensitive configuration (DB credentials, API keys) out of source code.

8.2. File Structure

templify_backend/

```
|—— configuration/
|   |—— cloudinary.js    # Cloudinary config
|   |—— db.js           # MongoDB connection
|—— controllers/
|   |—— fileController.js # All file-related logic
|—— models/
|   |—— File.js          # File schema
|   |—— User.js          # User schema
|—— routes/
|   |—— fileRoutes.js    # API route definitions
|—— index.js             # Entry point, server setup
|—— package.json         # Dependencies and scripts
|—— .env                 # Environment variables
```

Key Files:

index.js: Sets up Express server, connects to MongoDB, mounts routes.

fileRoutes.js: Defines API endpoints for file operations.

fileController.js: Contains business logic for uploading, listing, updating, and deleting files.

File.js/User.js: Mongoose schemas for files and users.

cloudinary.js/db.js: Configuration for Cloudinary and MongoDB.

8.3. API Endpoints & Functionality

Method	Endpoint	Description
POST	/api/files/upload	Upload a new file
GET	/api/files/list?email=...	List all files for a user
POST	/api/files/update	Update (replace) a file
POST	/api/files/delete	Delete a file
GET	/api/files/test	Test API status

Checks/creates the user in MongoDB.

Uploads the file to Cloudinary (publicly accessible).

Stores file metadata (filename, URL, userEmail, timestamp) in MongoDB.

User lists their files (history) by email.

Backend fetches all files associated with that email.

User updates a file by providing a new file, fileId, and email.

Backend deletes the old file from Cloudinary, uploads the new one, and updates the DB record.

User deletes a file by fileId and email.

Backend deletes the file from Cloudinary and removes the DB record.

8.4. Issues Faced During Development & Solutions

a. File Name in Cloudinary URL

Issue: Cloudinary was generating random public IDs, so URLs did not contain the original file name and extension.

Solution: Set the public_id in the upload call to uploads/<original_filename>, ensuring the URL contains the actual file name and extension.

b. Public Access to Files

Issue: Uploaded files were not accessible publicly (401 Unauthorized).

Solution: Ensured Cloudinary upload preset was set to "Public" and did not set type: "authenticated" or type: "private" in the upload call.

c. File Type Restriction

Issue: Users could upload any file type.

Solution: Used Multer's fileFilter to restrict uploads to only PDF, TXT, XLS, and DOC files.

d. User-File Association

Issue: Needed to associate files with users for personalized history.

Solution: Added a userEmail field to the File schema and ensured all file operations are filtered by user email.

e. Handling Duplicate File Names

Issue: Uploading a file with the same name could cause conflicts in Cloudinary.

Solution: Before uploading, the backend deletes any existing file with the same name in Cloudinary.

f. Environment Variables

Issue: Sensitive credentials (DB URI, Cloudinary keys) should not be hardcoded.

Solution: Used dotenv and a .env file to manage environment variables securely.

8.5. Security & Access

File Type Restriction: Only allows .pdf, .txt, .xls, .doc files.

Public File Access: Files are uploaded as public resources in Cloudinary for easy sharing.

User Isolation: Each file is associated with a user's email, so users can only manage their own files.

8.6. Conclusion

The Templify backend is a robust, scalable, and secure solution for user-based file management, leveraging modern web technologies and cloud services. It is well-suited for integration with any frontend (web, mobile, etc.) and can be extended for more advanced use cases as needed.

8.7. References

Cloudinary Documentation

Express.js Documentation

MongoDB Documentation

Multer Documentation

Node.js Documentation

8.9 API Documentation

Upload File:

POST /api/files/upload

Body: file, email (multipart/form-data)

List Files:

GET /api/files/list?email=...

Update File:

POST /api/files/update

Body: file, fileId, email (multipart/form-data)

Delete File:

POST /api/files/delete

Body: fileId, email (JSON)

Test API:

GET /api/files/test

CHAPTER 9

References and Appendices

9.1. Links to Key Documentation

Source code:

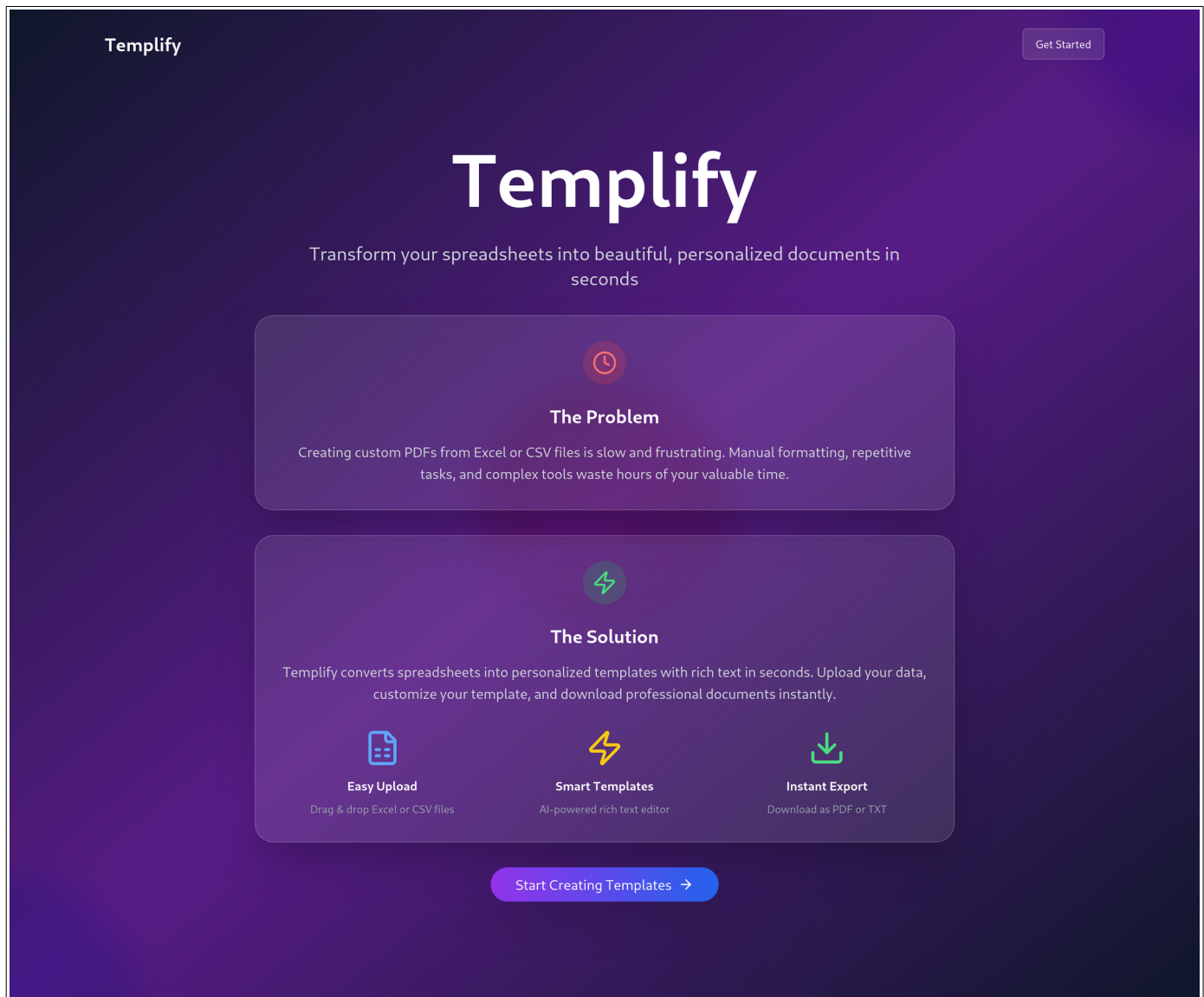
- **Frontend:** <https://github.com/abhishek-2k23/Templify>
- **backend:** https://github.com/abhishek-2k23/templify_backend

Official Documentation:

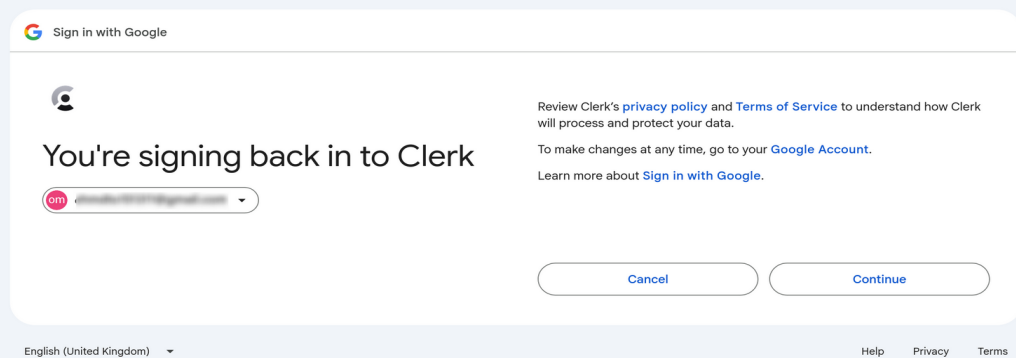
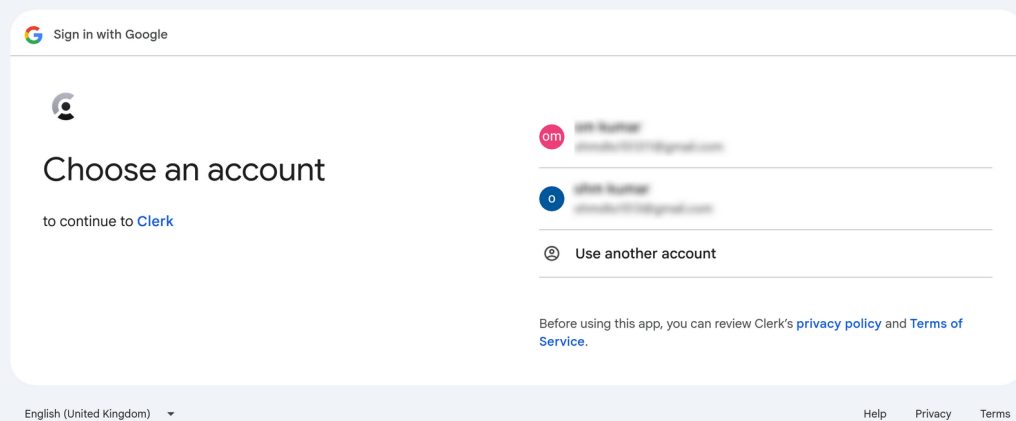
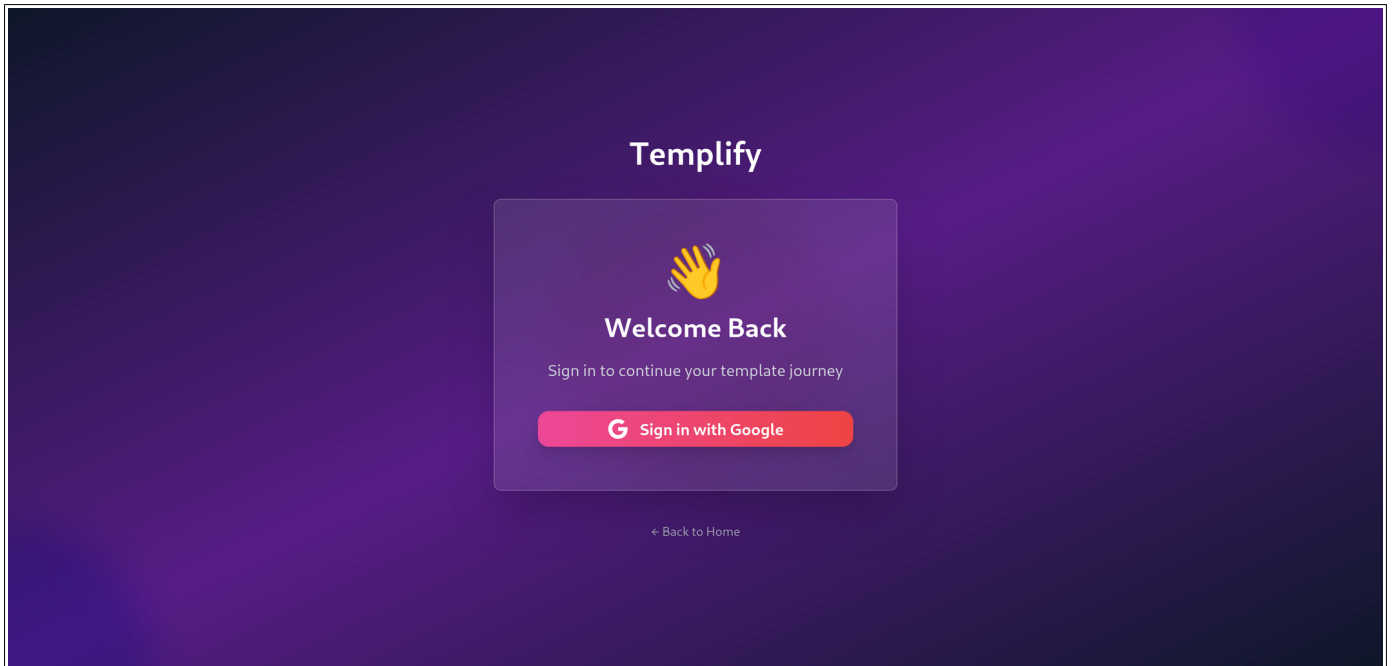
- React: <https://react.dev/>
- Vite: <https://vitejs.dev/>
- Tailwind CSS: <https://tailwindcss.com/>
- React Router: <https://reactrouter.com/>
- Clerk: <https://clerk.com/docs>
- TypeScript: <https://www.typescriptlang.org/docs/>
- ESLint: <https://eslint.org/docs/>
- Prettier: <https://prettier.io/docs/>
- react-hot-toast: <https://react-hot-toast.com/>

9.2. Screenshots of Key Frontend Features

1.Home Page:



2.Signin Page:



3.File Upload:

Templify

Home

History

om

Create Your Template

Upload your spreadsheet and design your personalized template

Upload Your Spreadsheet

Drag and drop or click to select Excel or CSV files (max 2MB)

Choose File

4.Create Template:

Templify

Home

History

om

Create Your Template

Upload your spreadsheet and design your personalized template

file_example_XLS_10.xls

8 columns detected

Available Headers:

@0

@First Name

@Last Name

@Gender

@Country

@Age

@Date

@Id

Design Your Template

PDF Header (optional):

file_example_XLS_10 data

This will appear at the top of your PDF.

Template Type:

Invitation

Helps the AI generate a better template.

Start typing your template... Use @ to insert headers from your spreadsheet, or click on the header blocks above

Download .txt

Download .pdf

5.History:

